UMESH KUMAR SAHOO

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EDUCATION

Silicon University, Bhubaneswar · B.Tech – Computer Science & Engineering — *CGPA: 9.72 / 10* *2023–2027*

Jawahar Navodaya Vidyalaya, Panikoili · Senior Secondary (PCMB) · CBSE — *91.2%* *2021*

Jawahar Navodaya Vidyalaya, Panikoili · Secondary · CBSE — *95.8%* *2019*

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), SQL, C

AI / ML / Gen AI: Machine Learning, NLP, RAG Pipelines, LLM Integration, Vector Embeddings, Whisper, Scikit-learn, XGBoost, Gradient Boosting, LSTM, RNN, ARIMA, Time Series Forecasting

Data Engineering: Pandas, NumPy, Feature Engineering, Data Preprocessing, Regression, Classification, Clustering, Statistical Modeling

Cloud & DevOps: GitHub Actions (CI/CD), Terraform, Infrastructure as Code (IaC), Linux

Web & Backend: React.js, Node.js, Express.js, MongoDB, Flask, REST APIs

Tools: TensorFlow, Keras, Jupyter Notebook, Postman, Matplotlib, Seaborn, Plotly, Git, GitHub

EXPERIENCE

DevOps Engineer Intern · **Ingenious Tech World** — *Jun 2025 – Jul 2025*

- Engineered **CI/CD pipelines via GitHub Actions** — automated build, test & deploy workflows, cutting manual deployment effort by **40%**.
- Provisioned scalable cloud infrastructure using **Terraform (IaC)**, ensuring environment consistency across dev and staging; applied distributed-systems principles for multi-region reliability.

MERN Stack Developer Intern · **Syllogistick** — *Jun 2024 – Jul 2024*

- Built production e-commerce platform: **MongoDB + Express.js + React.js + Node.js** with JWT auth and **10+ REST API** endpoints.
- Improved frontend responsiveness via React code-splitting; optimized MongoDB schema design to reduce query latency.

PROJECTS

RAG-Based Video Teaching Assistant ▸ *Python · Whisper · RAG · LLM · Vector Embeddings · Streamlit*

- Designed Gen AI Q&A system using **Retrieval-Augmented Generation (RAG)**: Whisper transcription → semantic embeddings → cosine similarity retrieval → timestamp-aware chunking.
- Implemented **confidence-threshold filtering** to suppress hallucinated responses, improving reliability across educational LLM workflows.

FreshGuard – Shelf-Life Prediction System ▸ *Python · Gradient Boosting · Scikit-learn · Flask · GitHub Actions CI/CD*

- Led end-to-end ML pipeline on **3,552 real-world records**; achieved **R^2=0.89** with Gradient Boosting — **31% improvement** over baseline after benchmarking 5+ regression models.
- Deployed trained model as **Flask REST API** with **GitHub Actions CI/CD**, enabling automated testing and cloud-ready continuous deployment.

Nifty-50 Stock Price Forecasting (ARIMAX · LSTM · RNN) ▸ *Python · TensorFlow · Keras · pmdarima · Statsmodels · LightGBM*

- Forecasted VWAP on **20-year ICICI Bank / Nifty-50 dataset**; engineered **32 lag features** via 3d/7d/30d rolling mean+std sliding windows across 5 financial indicators.
- Confirmed non-stationarity (**ADF stat: −2.69**); applied Box-Cox transformation + additive/multiplicative seasonal decomposition + ACF/PACF analysis before modelling.
- Benchmarked **AR(3), ARIMAX (auto_arima/pmdarima), LSTM (100 units), SimpleRNN**; LSTM outperformed RNN by **~3–4% MAE** with EarlyStopping (patience=21) on 80/20 split.

ACHIEVEMENTS & CERTIFICATIONS

- **Runner-Up** — SPEC2MODEL ML Challenge (Food Delivery Delay Prediction) | **2nd Place** — Algo Arena 2.0, Silicon University
- Google Generative AI (Intro) · IBM Introduction to AI · Job Ready Data Science · Active LeetCode problem-solver (DSA & Design Thinking)